Rutang Patel

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Profile

Computer Science student (CGPA 9.02) who builds **agentic AI systems in Python**. Built **Socrates**, a multi-agent research assistant on LangGraph with tool calling, structured outputs, and semantic long-term memory (Voyage AI + MongoDB Atlas Vector Search), and **Radix**, a FastAPI backend with 12+ secured endpoints. Hands-on with LLM APIs (OpenAI, Gemini, Groq) and a Transformer implemented from scratch, with top 12–26% global finishes in ML competitions. Eager to prototype AI agents fast and take the best experiments to production.

Education

G H Patel College of Engineering and Technology — Anand, Gujarat, India
B.Tech in Computer Science and Engineering (IoT) — ***CGPA: 9.02*** — *Expected May 2027*

Relevant coursework: Data Structures and Algorithms, Design and Analysis of Algorithms, Database Management Systems, Object-Oriented Programming, Operating Systems, Data Communication and Networking

Technical Skills

Languages: Python, C++, Java, JavaScript
Agentic AI and LLMs: LangGraph, LangChain, LLM Agents, Multi-Agent Workflows, Tool Calling, Structured Outputs, LLM APIs (OpenAI, Gemini, Groq), Semantic Search, Vector Search
Backend and Deployment: FastAPI, REST APIs, JWT Authentication, MongoDB Atlas, Render, Git
Machine Learning: TensorFlow, Keras, PyTorch, Scikit-learn, XGBoost, LightGBM, CatBoost, Optuna, NumPy, Pandas, Transformer Architectures, Ensemble Learning

Projects

Socrates – Agentic AI Research Assistant — GitHub
Python, LangGraph, LangChain, Tavily, MongoDB Atlas, Voyage AI

- **Built a multi-agent system** in LangGraph that routes each query to a memory, conversation, or web-research path using Planner, Researcher, Tool, and Summarizer nodes.
- Designed an **iterative research workflow**: the Planner splits a question into independent sub-questions via structured output, and the Researcher calls web-search tools until enough evidence is gathered for a sourced summary.
- Implemented **semantic long-term memory** with Voyage AI embeddings and MongoDB Atlas Vector Search, plus a common model factory that swaps between OpenAI, Gemini, and Groq.

Radix – Biometric P2P Payment System — GitHub
Python, FastAPI, MongoDB Atlas, DeepFace, ImageKit

- **Built and deployed** a P2P payment backend on Render with atomic MongoDB balance updates and rollback handling for failed transactions.
- Implemented **ML-based face authentication** using FaceNet512 embeddings, RetinaFace detection, anti-spoofing, and MongoDB Atlas vector search with a 0.8 cosine-similarity threshold.
- Shipped and secured **12+ production API endpoints** with JWT blacklisting, TTL-indexed token expiry, bcrypt-hashed credentials, and slowapi rate limiting.

picoGPT – GPT-Style Transformer from Scratch — GitHub
TensorFlow, Keras, NumPy

- **Implemented a decoder-only Transformer from scratch**: 6 decoder blocks, 8-head causal self-attention, learned positional embeddings, residual connections, layer normalization, and autoregressive generation.
- Built the **full training pipeline** on TinyStories (character-level tokenizer, tf.data sliding windows, GPU prefetching, Adam, checkpointing) and trained on an NVIDIA P100 to **92.48% next-character accuracy**.

ML Competitions

- **Top 12% globally** (1,324/10,631), Gridlock Hackathon demand forecasting: reached 0.9208 R^2 by finding **target-encoding leakage** and building a leak-free, time-based validation pipeline with CatBoost and LightGBM.
- **Top 26% globally** (776/3,022), Kaggle F1 Pit Stop Time Prediction: reached 0.95155 ROC AUC with XGBoost, LightGBM, and CatBoost, **Optuna-tuned and combined by out-of-fold stacking**.

Certifications

Machine Learning Specialization – Andrew Ng, DeepLearning.AI (Coursera)
Mathematics for Machine Learning and Data Science – DeepLearning.AI (Coursera)